

Krish Singal

CONTACT	Personal website: https://krishsingal.github.io LinkedIn: https://linkedin.com/in/krishsingal Email Address: ksingal@engineering.upenn.edu	
RESEARCH INTERESTS	I am broadly interested in theoretical computer science and machine learning. Recently, I have been thinking about problems related to efficient computation in machine learning systems.	
EDUCATION	University of Pennsylvania	2024–2029 (Expected)
	• PhD Candidate in Computer Science • Advised by Erik Waingarten and Sanjeev Khanna	
	Columbia University	2020–2024
	• Bachelor of Science in Computer Science, Magna Cum Laude	
	Dulles High School	2016–2020
	• Class Salutatorian	
INDUSTRY EXPERIENCE	Applied Scientist Intern, Amazon Web Services	Aug 2026 – Present
	Software Engineer Intern, Apple	May 2022 – Aug 2022
	Software Engineer Intern, Amazon Web Services	May 2021 – Aug 2021
SELECTED PUBLICATIONS	Inner Product Aware Quantization: Provably Fast, Accurate, and Adaptive Algorithms Krish Singal, Nathan White. Preprint	
	A Polynomial Space Lower Bound for Diameter Estimation in Dynamic Streams Sanjeev Khanna, Ashwin Padaki, Krish Singal, Erik Waingarten. Foundations of Computer Science (FOCS) 2025	
	MC²: Rigorous and Efficient Directed Greybox Fuzzing Abhishek Shah, Dongdong She, Samanway Sadhu, Krish Singal, Peter Coffman, Suman Jana. Conference on Computer and Communications Security (CCS) 2022 <i>Honorable Mention Best Paper Award</i>	
TALKS	A Polynomial Space Lower Bound for Diameter Estimation in Dynamic Streams • Foundations of Computer Science (FOCS), December 2025 • UPenn Theory Seminar, December 2025 On the Size and Complexity of Scrambles • Joint Mathematics Meetings (JMM), January 2024	
TEACHING	I have served as a teaching assistant for the following course offerings	

- Algorithms for Big Data, taught by [Anindya De](#) and [Erik Waingarten](#) (Summer 2026, Spring 2026, Fall 2025)
- Advanced Algorithms, taught by [Alexandr Andoni](#) (Spring 2024)
- Computational Complexity, taught by [Henry Yuen](#) (Fall 2023)
- Analysis of Algorithms, taught by [Christos Papadimitriou](#) (Spring 2023, Fall 2022)

SERVICE

Seminar Instructor

- Co-organizer of [Penn Theory Seminar](#) (2026 - Present)
- Co-led a seminar on the Analysis of Boolean Functions as part of the [Columbia Undergraduate Seminar in Theoretical Computer Science](#) (Fall 2023)

Conference sub-reviewer

- STACS 2026

COURSEWORK

Randomized Algorithms, Convex Optimization, Machine Learning Theory, Algorithms for Massive Data, Zero-Knowledge Proofs, Complexity Theory, Operating Systems, Distributed Systems, Quantum Computing.

SKILLS

Programming Languages: C/C++, Python, Java, LaTeX, GoLang, Rust
Languages: English (native, fluent), Hindi (full professional fluency), Spanish (full professional fluency)